

Can Mobility Subsidies Reduce Summer Travel Inequality?

Weather Shocks, Digital Exclusion, and Finite Supply in an AgentSociety² Synthetic City

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Abstract

Extreme heat and heavy rain increase the appeal of door-to-door mobility, but digital access, policy awareness, and finite vehicle supply determine who can actually obtain it. We use AgentSociety² to build a nine-zone synthetic city with three age and digital-access groups, three weather states, and finite ride-hailing supply. We test public, older-adult-targeted, and mixed coupons against a no-coupon baseline. The confirmatory policy experiment contains 200 agents, ten paired random seeds, and four policies. An evidence base, supported by 41 independent sources, contains 23 weather-transport records, nine older-adult hailing records, four promotion-rule records, and 42 community support records. These sources calibrate mechanism direction and parameter bounds; they do not turn synthetic outcomes into forecasts for Shanghai.

Under heavy rain, the final no-coupon baseline generated 64.8 ride-hailing requests and 1.6 failed requests on average, with 96.94% necessary-activity completion. Public, older-adult, and mixed coupons increased requests by 10.0, 3.0, and 8.0, respectively; all paired 95% confidence intervals excluded zero. Yet their paired effects on failures and necessary-activity completion all included zero. The public coupon cost the most (CNY 136.51 on average) without an identifiable completion gain. The targeted coupon cost only CNY 25.99 but redeemed just 5.0 coupons on average because awareness, participation, and digital-use frictions remained. In the group experiment, heavy-rain requests per 100 people declined from 51.8 among ages 18–39 to zero among older adults with neither digital ability nor assistance. This is not absent need; it is demand that never enters the platform. A supply contraction from 36 to 24 vehicles increased the share of seeds experiencing competition from 60% to 100%, showing that capacity thresholds govern whether induced demand becomes crowd-out.

The central result is that coupons change visible demand but do not, by themselves, remove access barriers. When unconstrained public transit provides a fallback, aggregate completion can conceal distributional inequality. Policy evaluation should therefore report who enters the application funnel, who receives and redeems support, and who completes necessary activities under finite supply. Price subsidies should be combined with telephone and offline channels, assisted booking, priority for necessary trips, and supply management.

Keywords: agent society; transport equity; ride-hailing subsidy; extreme weather; digital divide; induced demand; urban computing

Main finding. In this synthetic city, coupons reliably induced ride-hailing demand during heavy rain but did not reliably improve necessary-activity completion. The most digitally excluded older adults could not enter either the coupon or hailing funnel. A price instrument therefore needs non-digital access and capacity governance.

1 Introduction

Summer weather shocks alter the monetary, temporal, and health costs of urban travel. Extreme heat increases outdoor exposure costs, while heavy rain suppresses walking and shifts travelers toward transit and door-to-door modes. Ride-hailing capacity, however, cannot expand instantaneously. Transport equity research consequently asks not only whether average accessibility improves, but also how benefits, burdens, and opportunities are distributed across groups[4, 5]. Ride-hailing can additionally create vehicle miles,

deadheading, and congestion externalities[6]. Older adults face a layered constraint: smartphones, payment, app procedures, awareness, and confidence can each block access[9, 10].

Coupon programs appear to offer a simple response. Public coupons maximize reach, targeted coupons prioritize disadvantaged users, and mixed programs attempt a compromise. Yet issuing a coupon does not guarantee use. If discovery and redemption occur only in an app, digitally excluded users can leave the process at awareness, participation, award, redemption, or booking. Meanwhile, subsidized trips compete for finite vehicles and may crowd out trips that would otherwise have succeeded. Static policy evaluation rarely represents daily schedules, exposure, platform use, road conditions, and policy behavior in one feedback system.

Generative-agent platforms make this problem executable[2, 3, 1]. We encode the policy as a counterfactual simulation and ask:

1. How do extreme heat and heavy rain alter mode choice, hourly flows, and exposure across groups?
2. Do coupons improve necessary-activity completion, or mainly induce requests and supply competition?
3. How do digital ability and family or community assistance change policy access for older adults?

This paper contributes an auditable chain from public evidence to a synthetic city, multi-agent feedback, and paired counterfactuals. It explicitly models a four-stage coupon funnel and finite ride-hailing supply. It also separates statistical confirmation from mechanism demonstrations and scalability checks, so that a single API run or an unfinished high-performance framework is not presented as causal evidence.

2 Mechanisms and hypotheses

2.1 Weather, substitution, and exposure

Rain changes the temporal and spatial pattern of taxi demand[7]; extreme heat also alters the relative generalized cost of transit and active travel[8]. Let the generalized cost of mode m for person i at time t be

$$G_{itm} = \alpha T_{itm} + \beta C_{itm} + \gamma R_{it}(w, m) + \delta A_{it}(m), \quad (1)$$

where T is time, C monetary cost, R weather exposure risk, and A access friction. A coupon lowers C , while digital exclusion raises A . Finite supply feeds back through waiting and failure, increasing T . A subsidy can therefore increase requests without increasing completed activities.

2.2 The policy funnel

We factor coupon use into

$$P(\text{redeem}) = P(R)P(P | R)P(A | P)P(U | A), \quad (2)$$

where R , P , A , and U indicate reach, participation, award, and use. Targeting older adults does not automatically increase any conditional probability. Shanghai’s one-click and 95128 telephone hailing programs illustrate why a non-app channel is a substantive part of mobility policy rather than a minor interface choice[13, 14, 15].

2.3 Testable expectations

We expect: H1, heat and rain increase the relative appeal of ride-hailing; H2, coupons induce requests; H3, the public coupon produces more redemption and expenditure than targeted support; H4, induced demand creates failures when vehicles are scarce; H5, an older-adult coupon raises necessary-activity completion for its target group; and H6, digital ability and assistance strongly moderate access. We report unsupported expectations rather than treating them as confirmed.

3 Data and code provenance

3.1 Audit of all repository branches

We froze and inspected all four remote branches, not only the default branch. Figure 1 summarizes their roles and Table 1 records the exact commits. The `Yoyo_database` evidence work was merged into `main`; `Yoyo` added policy components and scale-oriented code; `experiment` contains the final behavior confirmation, supply stress tests, and 200/1000-agent API reports. Because later commits updated behavior, the main results use the latest `final_coupon_*` files from `experiment`, not values copied from an older competition card.

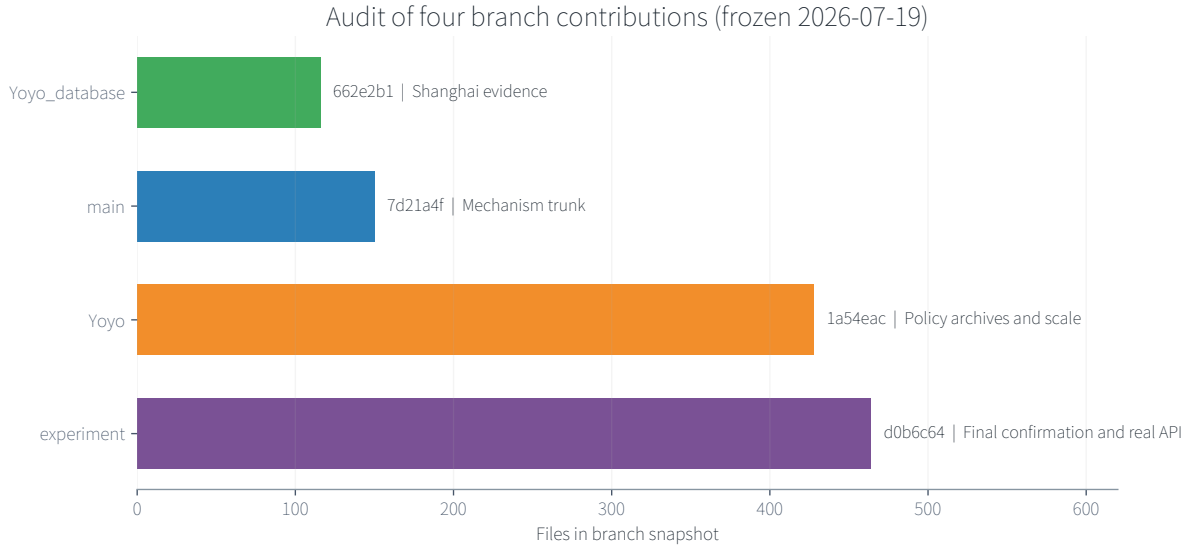


Figure 1: Evidence, mechanisms, and experiments across all four GitHub branches.

Table 1: Frozen repository provenance

Branch	Commit (first 12 characters)	Role in this paper
main	7d21a4f81cd8	Core city, weather, transport, and evidence framework
Yoyo_database	662e2b1c7051	Historical origin of the public evidence base
Yoyo	1a54eacece5f	Coupon module, result pipeline, frozen 10k/100k framework
experiment	d0b6c6498fc3	Final outputs, supply stress, and real-API experiments

3.2 Public evidence

The source registry contains 1,744 rows, but a registry row is not a verified fact. The core model uses four collections linked to 41 independent sources: 23 weather-transport records (22 high and one medium confidence), nine older-adult hailing-access records (eight high and one medium), four promotion-rule records (three high and one medium), and 42 subdistrict support records (ten high and 32 medium). A further 73 candidates were excluded for duplication, relevance, or quality. Figure 2 shows the core evidence structure.

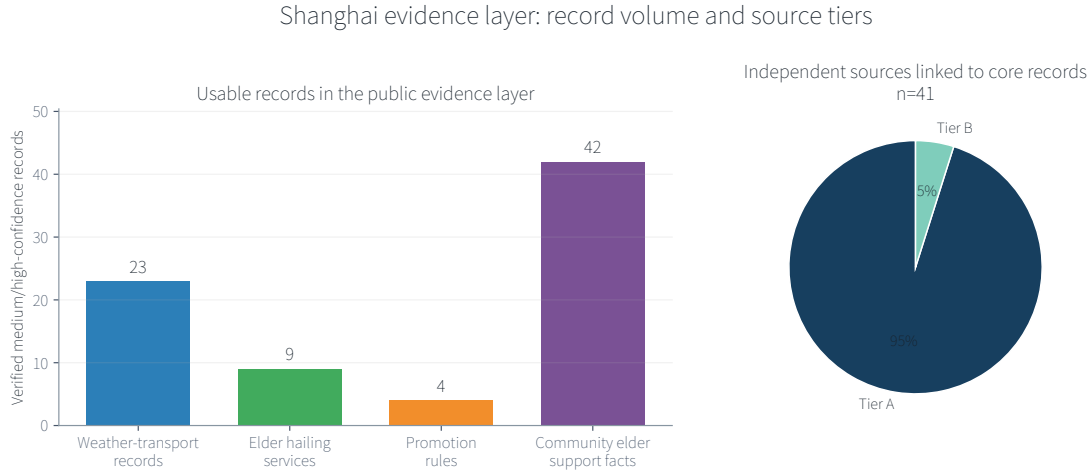


Figure 2: Core public evidence by topic and confidence. It constrains mechanisms and bounds; it does not directly validate synthetic outcome levels.

The mapping table separates observed, derived, and assumed quantities. Telephone access has high-confidence policy evidence. Community assistance is derived from local support records. Road-capacity multipliers, awareness probabilities, claim probabilities, and family assistance rates remain scenario assumptions. We therefore interpret results as mechanism tests and policy stress tests, not point forecasts for a real city.

4 Model and experimental design

4.1 Nine-zone synthetic city

The city consists of nine contiguous functional zones: an employment core, mixed and old inner-city areas, an industrial new town, a comprehensive sub-center, outer residential areas, and a remote transit-weak zone. A transport network supports walking, bus, metro, and ride-hailing. Weather changes mode utility, exposure risk, and road capacity. Figures 3 and 4 show the spatial structure and executable feedback loop.

Each agent has age, digital ability, possible family assistance, home and activity locations, necessary and discretionary activities, and an hourly schedule. Agents are grouped into ages 18–39, 40–59, and 60+, with older adults further divided into digital self-service, family proxy, and neither digital access nor assistance. Agents form an activity plan and then select a mode under current weather, road, price, coupon, and personal constraints. Road events update later decisions, so agents share an evolving environment rather than completing isolated conversations.

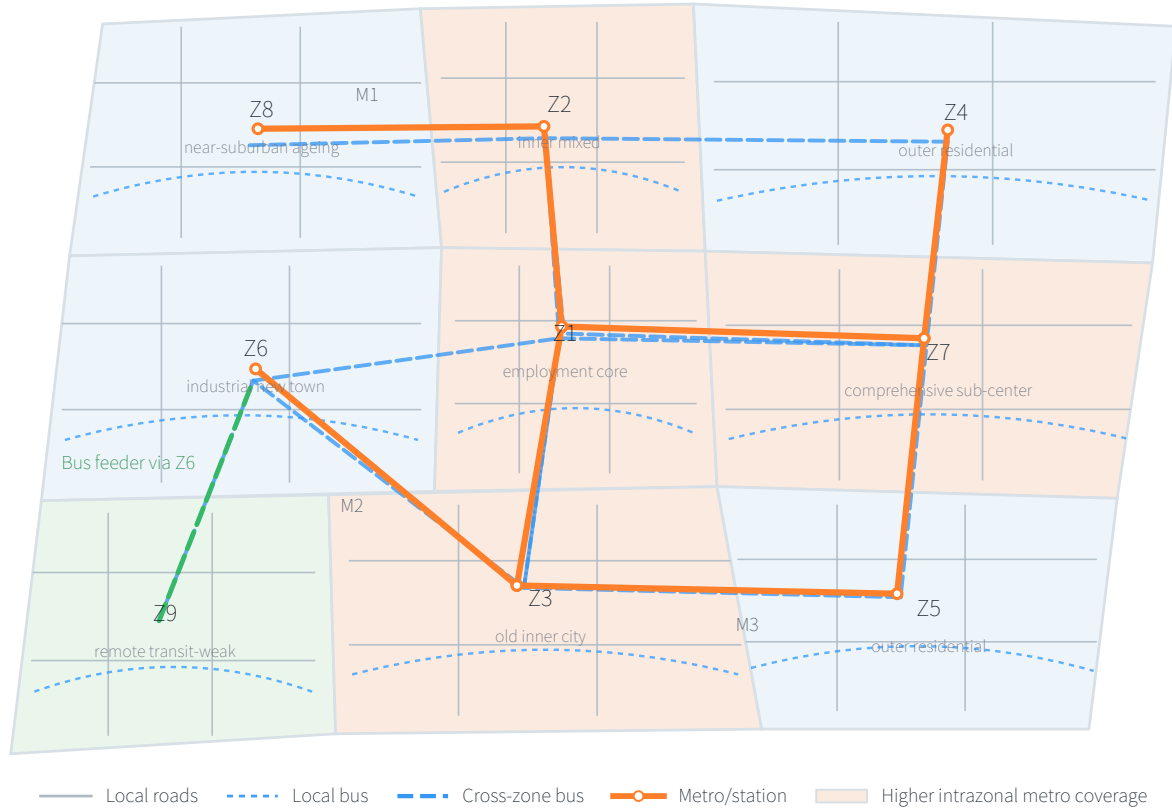
4.2 Weather, supply, and policies

Weather states are W0 normal, W1 extreme heat, and W2 heavy rain. Ride-hailing vehicles are finite and become available again only after a trip. A failed request can trigger bus, metro, or cancellation fallback. Four policies are compared:

- C0: no coupon;
- C1: public limited coupon with broad reach and competition for 40 awards;
- C2: older-adult limited coupon with targeted allocation but access friction;
- C3: mixed public and older-adult allocation.

Nominal awards are equal, but redemption and expenditure differ, so these are not equal-budget interventions. Each weather-policy cell in the main experiment contains 200 agents and ten paired random seeds.

Nine-zone multimodal network (redrawn from repository configuration)



M1 Z8-Z2-Z1-Z7-Z4 M2 Z6-Z3-Z1 M3 Z3-Z5-Z7 Intrazonal metro: Z1 / Z2 / Z3 / Z7

Figure 3: Nine-zone geography, multimodal routes, and intrazonal services. M1/M2/M3, cross-zone bus order, and the Z9 feeder are redrawn from `multimodal_transport_network.json`; pale orange denotes higher intrazonal metro coverage in Z1/Z2/Z3/Z7.

Research chain from public evidence to auditable counterfactuals

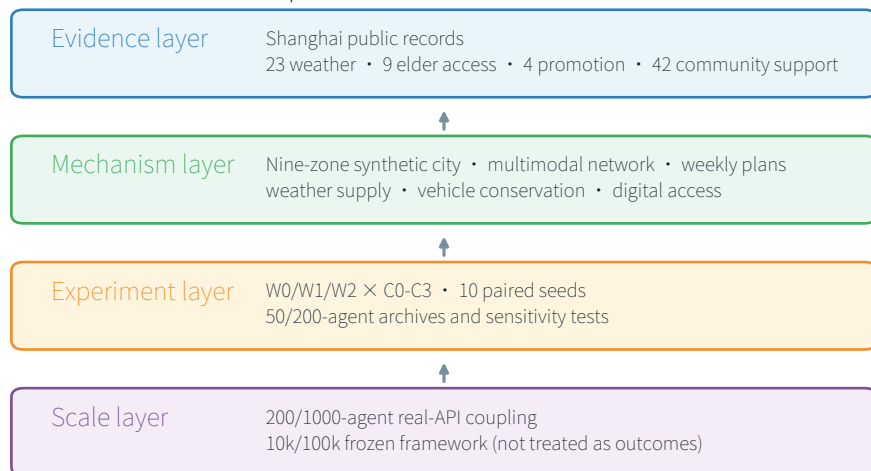


Figure 4: Agent-environment-policy feedback from public evidence to paired counterfactuals.

For outcome Y , the paired estimate is

$$\bar{d} = \frac{1}{10} \sum_{s=1}^{10} (Y_{s,p} - Y_{s,C0}), \quad CI_{95\%} = \bar{d} \pm t_{0.975,9} \frac{s_d}{\sqrt{10}}. \quad (3)$$

4.3 Experimental hierarchy and inference boundary

Table 2: Experimental layers and inferential roles

Layer	Scale/repeats	Computation	Supported interpretation
Detailed tide trace	50, one seed	Rule/frozen	Spatial and hourly mechanism illustration only
Policy confirmation	200, ten seeds	Frozen decisions	Paired means and 95% confidence intervals; main inference
Supply stress	200, multiple seeds	36 vs. 24 vehicles	Capacity threshold for competition and failure
Real API validation	200/1000, one seed	LLM API	Call-chain, coupling, and scale feasibility
HPC framework	10k/100k	Frozen code	No accepted outcome; excluded from empirical claims

Outcomes include requests, successes and failures, the reach-participation-award-redemption funnel, subsidy expenditure, necessary-activity completion, mode shares, hourly origin-destination flows, and group weather risk. Necessary-activity completion is the principal welfare outcome, but it is jointly determined by fallback transit and capacity assumptions. It must therefore be read together with access and failure metrics.

5 Results

5.1 Weather changes mode structure and urban tides

Without coupons, normal-weather mode shares were approximately 1.2% walking, 57.0% bus, 35.4% metro, and 6.2% ride-hailing. Ride-hailing rose to about 7.9% under heat. Heavy rain produced a larger substitution: bus fell to about 39.9%, metro rose to 42.5%, ride-hailing to 16.9%, and walking fell to roughly 0.5% (Figure 5). This supports H1 and matches the empirical direction that rain changes taxi demand[7].

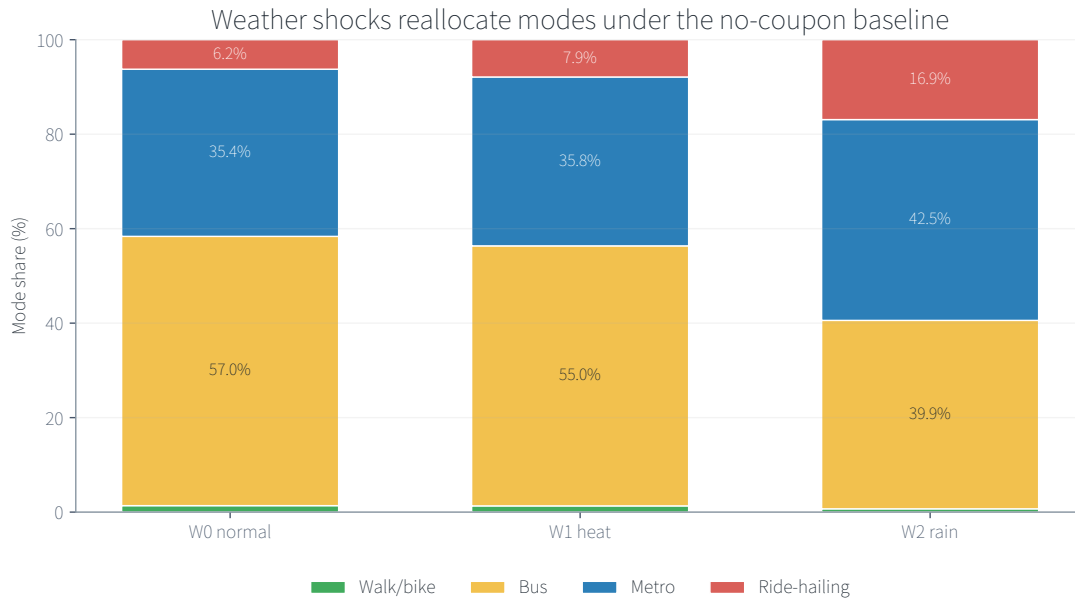


Figure 5: Mode shares by weather under the no-coupon baseline (200 agents, ten-seed means).

The detailed 50-agent trace in Figure 6 illustrates an outbound morning peak from residential zones toward employment, education, and transport nodes and an evening return. Heavy rain suppresses walking and shifts flows toward public transport and ride-hailing. Because this figure is a single-seed structural diagnostic, it is not an estimate of real passenger counts.

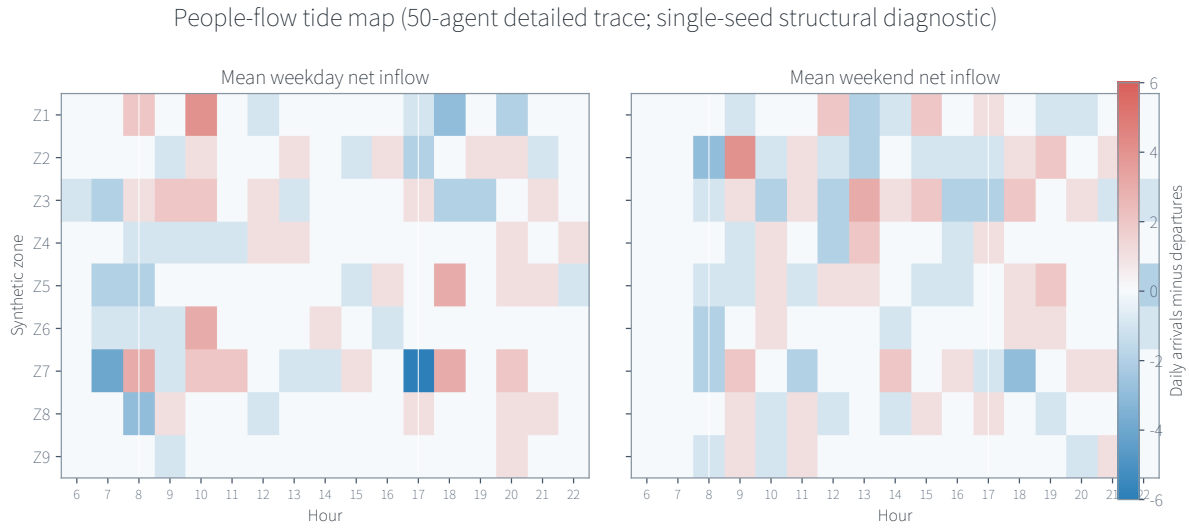


Figure 6: Hourly travel tide and heavy-rain origin-destination heatmap in the nine-zone city (50-agent diagnostic trace).

5.2 Coupons expand demand before they improve completion

Under heavy rain, average reach for C1, C2, and C3 was 191.4, 45.4, and 195.6 agents; participation was 128.6, 0, and 128.6; all three nominally awarded 40 coupons, but only 18.6, 5.0, and 14.2 were redeemed (Figure 7). C2's zero participation reflects direct targeted allocation rather than a public entry stage; it does not mean nobody encountered the policy. The consequential gap is between nominal award and actual use.

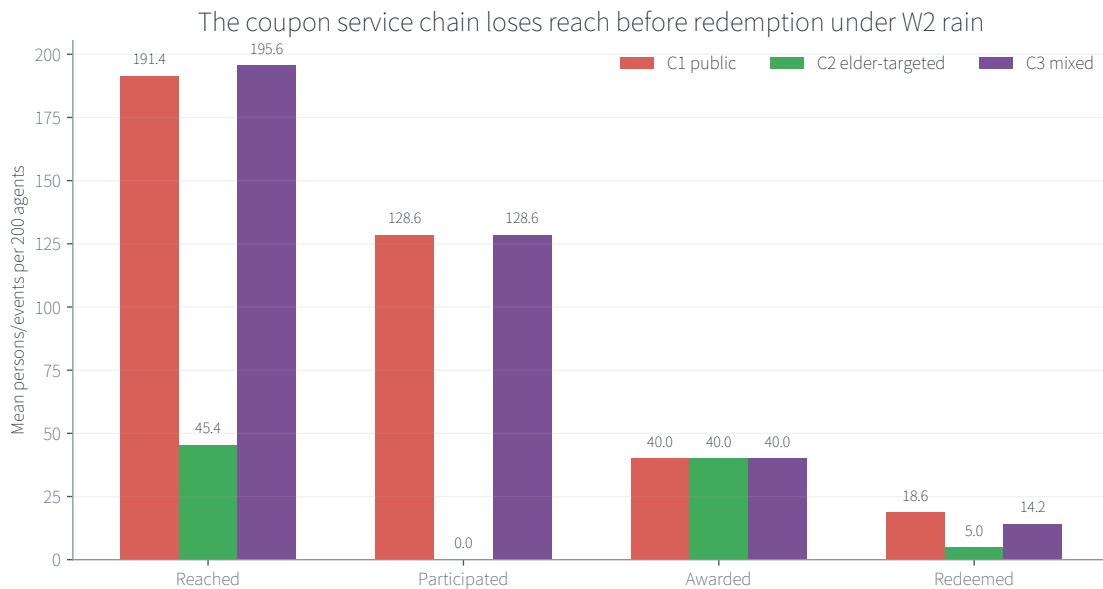


Figure 7: Coupon funnel under heavy rain (200 agents, ten-seed means).

Table 3 reports levels and paired effects. All coupons reliably induced requests: C1 +10.0 [7.22, 12.78], C2 +3.0 [1.88, 4.12], and C3 +8.0 [5.87, 10.13]. H2 is supported. In contrast, all confidence intervals for

failed requests and necessary-activity completion included zero. H5 is not supported, and H4 is not identified at the main capacity.

Table 3: Main policy results under W2 heavy rain (200 agents, ten paired seeds)

Outcome	C0 none	C1 public	C2 older	C3 mixed
Requests, mean	64.8	74.8	67.8	72.8
Failed requests, mean	1.6	2.7	1.8	3.0
Necessary completion, %	96.939	96.938	96.873	96.938
Coupons redeemed, mean	0.0	18.6	5.0	14.2
Subsidy, CNY	0.00	136.51	25.99	99.82
Paired request effect	-	+10.0	+3.0	+8.0
Request effect 95% CI	-	[7.22,12.78]	[1.88,4.12]	[5.87,10.13]
Paired failure effect	-	+1.1	+0.2	+1.4
Completion effect, pp	-	-0.001	-0.067	-0.001

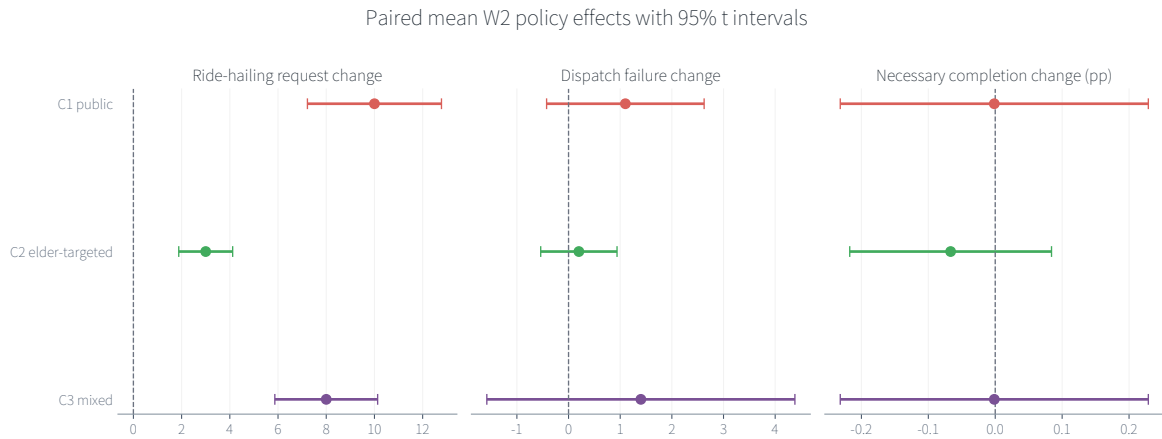


Figure 8: Paired policy effects relative to C0, with 95% confidence intervals. Completion effects are in percentage points.

The public coupon had the greatest expenditure and request increase but no corresponding completion gain. The targeted coupon spent least, while low redemption exposed access friction. Figure 9 places fiscal cost, redemption, and completion in the same comparison. H3 is supported. C2 should not automatically be labeled “optimal,” however, because the policies are not equal-budget and aggregate completion is insensitive to some distributional changes.

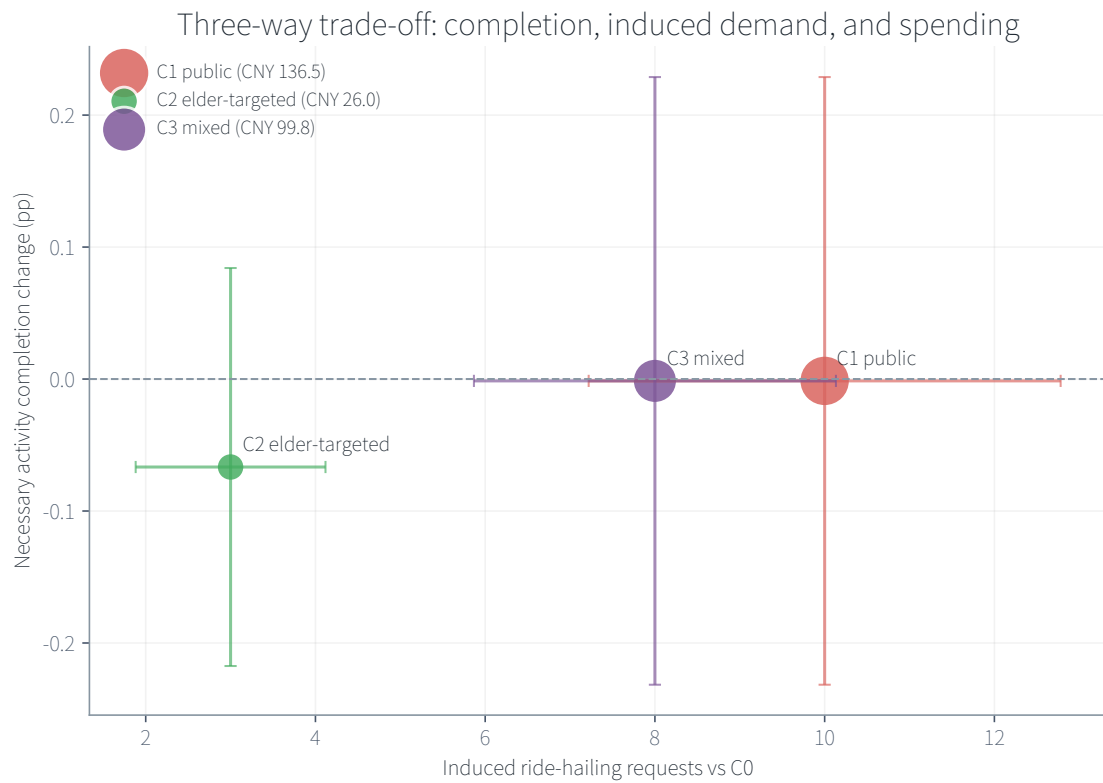


Figure 9: Fiscal expenditure, coupon redemption, and necessary-activity completion under heavy rain.

5.3 The digital divide creates invisible demand

In the joint older-adult confirmation, heavy-rain requests per 100 people fell from 51.8 among ages 18–39 and 31.5 among ages 40–59 to 7.3 for older digital self-users and 3.6 for older adults using family assistance. Older adults with neither digital ability nor help made zero requests (Figure 10). Zero platform requests are not evidence of zero mobility need. The group retained necessary activities but could not create an app-based request, so platform data systematically understate latent demand. This supports H6.

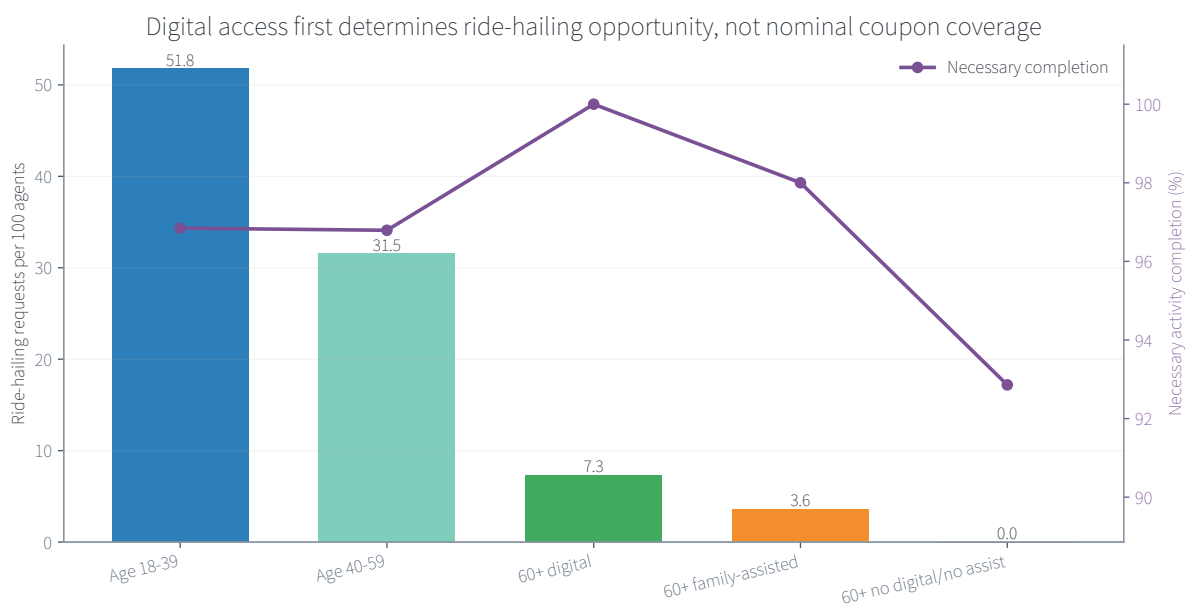


Figure 10: Age-digital access differences in heavy-rain requests and necessary-activity completion.

Aggregate completion did not collapse for the digitally excluded group because the model permits bus, metro, and schedule fallback. This reveals a measurement trap: binary completion treats dependence on help, longer waiting, unsuitable modes, and inability to enter the platform as equivalent to frictionless completion. Telephone, one-click, and proxy services address precisely this gap[13, 14].

The heat experiment also shows that older groups bear greater heat-risk burden despite fewer outdoor minutes (Figure 11). Age inequality is thus not only about mode access; the same environmental exposure can carry unequal health costs.

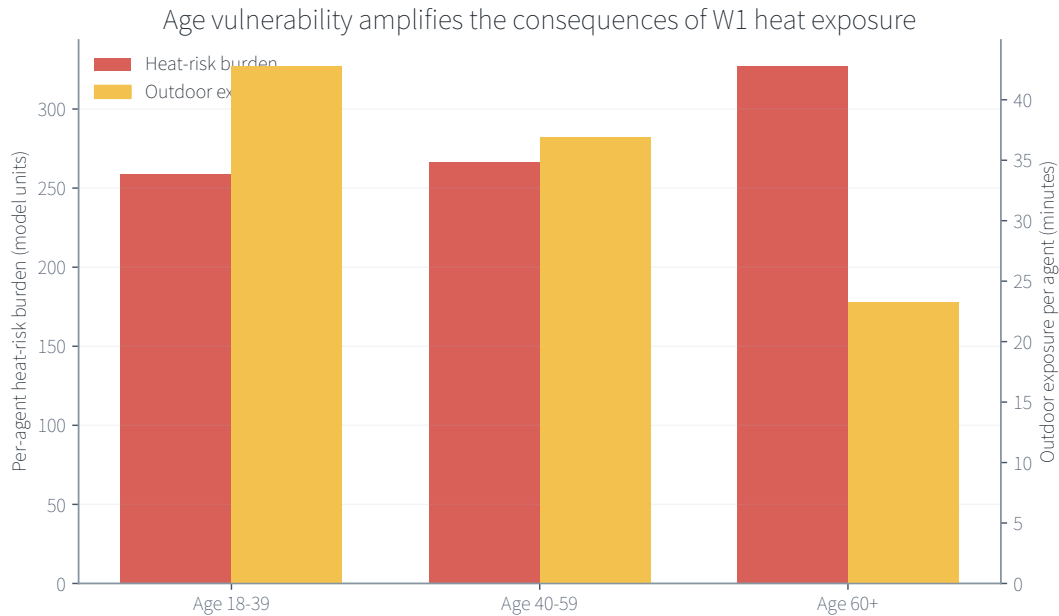


Figure 11: Outdoor exposure and heat-risk burden by age group under extreme heat.

5.4 Finite supply governs when externalities appear

In the 36-vehicle mechanism stress test, heavy rain produced 56.4 requests and 1.2 failures on average; 60% of seeds experienced competition and 10% experienced marked collapse. At 24 vehicles, the heavy-rain failure rate rose to about 6.7% and all seeds experienced competition (Figure 12). The main experiment's near-zero completion effect does not imply no congestion risk. The system had not consistently crossed the capacity threshold. H4 receives mechanism support under supply contraction.

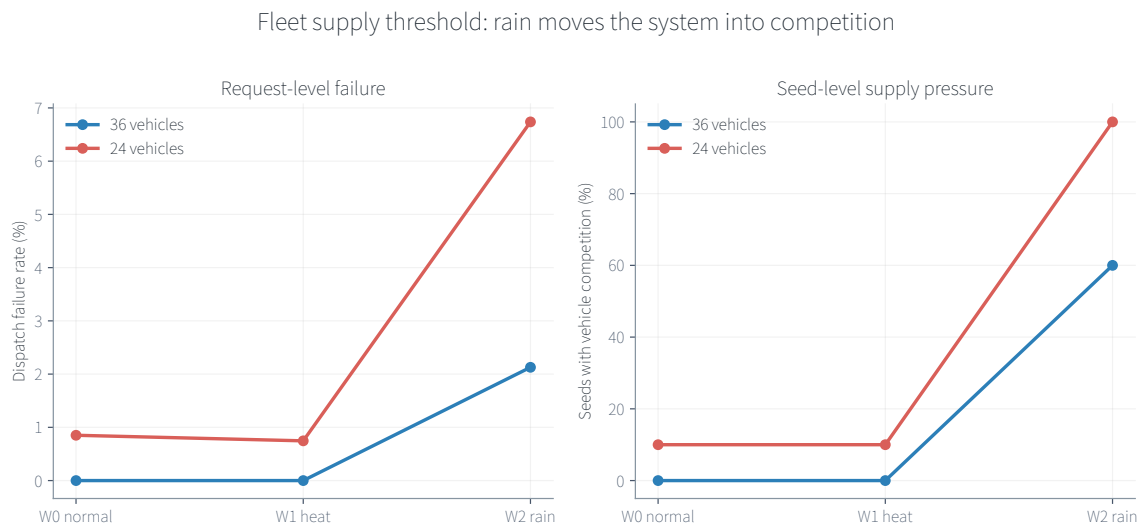


Figure 12: Request failure and share of competing seeds with 36 versus 24 vehicles.

5.5 Real-API coupling and scalability

Single-seed real-LLM runs tested whether behavior could operate through actual model calls rather than only frozen decisions. The 200-agent run produced 395 successful travel decisions, 76 ride-hailing road events, 230 decisions affected by prior environmental changes, 58.2% coupling coverage, and 554 influence edges. The 1000-agent run produced 1,996 travel calls and 622 contribution calls, 396 road events, 1,683 affected decisions, 84.3% coupling coverage, and 13,652 influence edges. It consumed approximately 1.21 million tokens and 1,404 seconds. Both runs reported zero API failures.

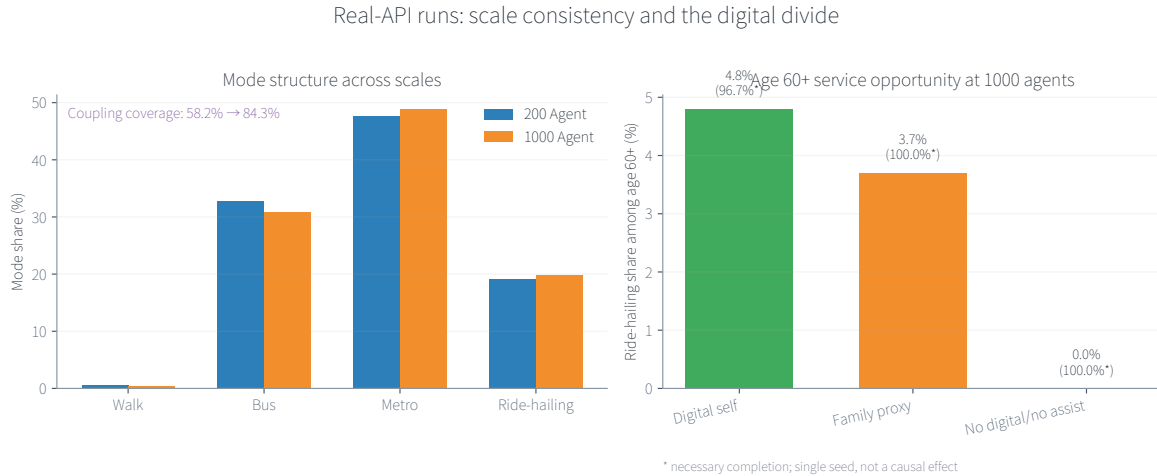


Figure 13: Calls, state coupling, and computational cost from 200 to 1000 agents in real-API runs.

In the single 1000-agent heavy-rain run, mode shares were 0.4% walk, 30.8% bus, 48.9% metro, and 19.8% ride-hailing. Ride-hailing shares were 33.2% for ages 18–39, 11.9% for ages 40–59, and 3.5% for ages 60+. These observations validate the execution and feedback chain, not statistical effects, because they use one seed and can vary with prompts and model versions. The 10k/100k code is a frozen framework with no accepted outcome and is excluded from empirical results.

6 Discussion and urban-science implications

6.1 Aggregate completion is not a sufficient equity metric

High completion in the main experiment is partly produced by public-transit fallback and the absence of explicit bus and metro capacity. Policy differences are absorbed by alternative modes. A fair evaluation therefore needs three layers:

1. **Opportunity:** reach, ability to participate, and availability of a digital or offline entry point;
2. **Service:** award, redemption, booking success, and waiting;
3. **Outcome:** necessary-activity completion and the distribution of weather risk and fiscal benefit.

An outcome-only measure counts a trip completed through dependency, detour, or discomfort as equivalent to an unimpeded trip.

6.2 From coupons to a price-access-capacity bundle

The public coupon’s strong demand response shows that price changes behavior, but high-peak demand can convert expenditure into competition. Targeting reduces expenditure yet does not remove funnel friction. A coherent policy bundle should retain telephone, one-click kiosk, community desk, and family-proxy access; link coupon release to supply, time, and critical origin-destination pairs; and prioritize medical, care, and emergency activities rather than allocate solely by app speed.

6.3 The value of executable social science

The system’s value is not an exact replica of one city. It turns theoretical claims into programs that can fail. Without digital ability in the decision process, older-adult exclusion cannot emerge. With unlimited vehicles, congestion externalities cannot emerge. Without state write-back, social feedback becomes parallel independent dialogue. These assumptions are inspectable and replaceable, and counterfactuals can be compared using the same random seed.

7 AI collaboration and reproducibility

7.1 Human-AI division of labor

AI supported four activities: extracting candidate facts from public pages and papers; generating or checking activity and travel decisions; assisting tests, data pipelines, and visualization; and identifying inconsistent reporting. Human researchers retained responsibility for the research question, evidence inclusion, parameter classification, counterfactual design, anomaly audit, and final interpretation. Every empirical claim required a source URL and confidence grade. Generated content was never automatically promoted to an observed fact.

7.2 Reproducibility audit

The analysis copied raw CSV files from frozen commits without overwriting the source repository. A dedicated script recomputed group means, paired differences, and Student- t intervals, producing machine-readable `derived_metrics.csv` and `analysis_audit.json`. After regeneration, the repository’s competition-report builder passed its `--check` consistency test, and targeted tests for the 1000-agent result pipeline passed. The audit found that an older report disagreed with the latest final behavior output; this paper therefore recomputes the values rather than copying stale tables.

The replication sequence is: check out all four commits; retrieve large-file objects; run result consistency checks; read final per-seed CSV files; run the bilingual plotting script; compile with XeLaTeX; and rasterize every PDF page for visual inspection. Repository: https://github.com/JennyXi/UrbanCup_2026_Jenny-Yoyo.

8 Limitations

1. **External validity.** The nine-zone city is theoretical. Evidence constrains mechanisms rather than estimates every parameter, so outcome levels cannot be extrapolated to Shanghai.
2. **Statistical power.** The main experiment has ten paired seeds and rare failures. Wide intervals mean that “not identified” is not equivalent to “no effect.”
3. **Capacity.** Bus and metro are not capacity constrained like ride-hailing. Fallback may overstate resilience and suppress measured inequality.
4. **Policy comparability.** Nominal awards, not budgets, are equal. Fiscal-efficiency comparisons are descriptive.
5. **Behavioral parameters.** Awareness, participation, family assistance, and capacity multipliers remain assumptions requiring survey or platform calibration.
6. **LLM stability.** Real-API outputs depend on model, prompt, temperature, and service version and use only one seed.
7. **Welfare measurement.** Completion omits waiting, dependency, discomfort, abandonment, and health loss. Future work should build group-specific welfare measures.

9 Conclusion

Can mobility subsidies reduce summer travel inequality? The answer is conditional. Under heavy rain, every coupon reliably increased ride-hailing requests, but none reliably improved necessary-activity completion. Targeted support lowered expenditure, yet some intended beneficiaries still exited before booking because of digital and assistance barriers. When vehicle supply contracted, induced demand quickly became competition and failure.

The policy object is therefore not a coupon but an accessibility chain: information must arrive, application must be possible, service must be supplied, and necessary activities must be completed with acceptable risk and dignity. Only a combination of price support, telephone and offline entry, assisted booking, priority for necessary trips, and dynamic supply management can plausibly convert more platform orders into greater mobility equity.

Data and materials availability

Raw experimental data, code, and branch history are available in the project GitHub repository. The accompanying artifact includes independently compilable Chinese and English LaTeX sources, bilingual vector figures, the analysis script, derived metrics, and final PDFs. Public-evidence URLs and record mappings are retained in the source registry. Reproducing third-party LLM calls requires appropriate credentials and version configuration.

A Key parameter classifications

Table 4: Key mechanisms and evidence status

Module	Example	Status	Interpretation
Weather-transport	Rain lowers road capacity and walking appeal	Direction observed; magnitude assumed	23 records support direction; multipliers such as 0.7/0.85/1.0 are not estimates
Older access	95128 telephone and one-click hailing	Observed/high confidence	Official Shanghai pages directly support non-app entry
Community assistance	Community or family proxy	Derived/medium confidence	Availability is bounded using 42 local support records
Coupon behavior	Awareness, participation, award, redemption	Partly assumed	Rules support policy form; behavioral probabilities need calibration
Ride-hailing supply	36-vehicle stress and 24-vehicle contraction	Scenario	Identifies a capacity threshold, not a real fleet size
Public transport	Fallback after request failure	Structural assumption	Bus and metro capacity is not explicit, an important cause of high completion

B Interpretation rules

We apply the following constraints: an interval crossing zero is not described as a significant improvement or deterioration; a single seed is a mechanism validation; frozen data supersede a conflicting old report after

recomputation; framework code without accepted output supports no performance or social claim; and a source registry count is never substituted for the number of independently supported facts.

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